

Samir Mohammed

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EDUCATION

University of Waterloo

September 2022 – Present

- **BMath (Honours)** in Computational Mathematics & Minors in Computer Science & Economics
- **Certifications:** CFA Level I (November 2025), Microsoft Azure AZ-305 (Solutions Architect Expert)
- **President's Scholarship of Distinction:** Achieved 95% or higher admission average

EXPERIENCE

Data & Strategy Intern

May 2026 – Present

Capital One

Toronto, ON

- Developing Capital One Canada's first customer referral program for upmarket credit card acquisition
- Supporting the modernization of credit risk modeling workflows by transitioning from GBM-based models to dynamic linear models (DLMs)

Data & Strategy Intern

September 2025 – December 2025

Capital One

Toronto, ON

- Proposed novel growth strategy for the customer-initiated Credit Limit Increase Program, expanding credit access while preserving portfolio risk thresholds
- Built Python simulation and valuation models to evaluate portfolio-level profitability, test policy scenarios, and analyze risk sensitivity under varying macroeconomic conditions
- Analyzed historical request data to identify an approval policy projected to lift rates from under 1% to over 9% and expand access for thousands of customers within portfolio risk thresholds
- Presented analytical findings to senior stakeholders across Product, Risk, and Finance, with the recommended policy approved for rollout in 2026

Business Analyst Intern

January 2025 – May 2025

Communitech

Kitchener, ON

- Redesigned the due diligence workflow for a federal funding program (ElevateIP), standardizing eligibility evaluation and improving intake efficiency across 450+ startups
- Automated the cost-audit pipeline using Apps Script and structured validation logic, reducing manual review time by 95% across audit cycles

Data Science Intern

May 2024 – August 2024

NeoStats

Bangalore, IN

- Developed PySpark ML models in Azure Databricks to optimize in-app cross-selling for a retail client, driving a 12% conversion lift validated through A/B testing
- Built ETL pipelines in Microsoft Fabric for a financial services client to support analytics and modeling workflows

PROJECTS

Portfolio Engine | *Next.js, TypeScript, Recharts, PapaParse, Zod*

- Built a browser-based portfolio analytics dashboard that parses brokerage holdings CSVs and surfaces allocation, exposure, and P/L insights across sector, industry, geography, and asset class
- Implemented in-browser parsing with schema validation and decimal.js for precise financial math, including sector-weighted ETF analysis to surface true underlying market exposure

Review Rating Prediction using NLP and Deep Learning | *TensorFlow, RoBERTa, LightGBM, scikit-learn*

- Built stacked ensemble combining RoBERTa transformer embeddings with LightGBM for multi-class rating prediction
- Trained on rented GPU infrastructure with hyperparameter tuning and sentiment-derived feature engineering

Blackjack Monte Carlo Simulator | *TypeScript, PostgreSQL, Supabase, Zod*

- Built a full-stack simulation platform with configurable rule sets, betting systems, and persisted simulation tracking
- Implemented simulation logic to compute risk-of-ruin, expected value, bankroll volatility, and strategy performance

TECHNICAL SKILLS

Languages: Python (NumPy, pandas, scikit-learn, PySpark), SQL, TypeScript, R, C++

Tools & Platforms: Azure Databricks, Microsoft Fabric, Snowflake, PostgreSQL, Supabase, Git, Jupyter, Unix Shell